The sharp Berry–Esseen bound for independent arrays with small Lyapunov ratio

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6 October 2026 Version 0.7.0

Abstract

Every finite independent centered array with positive total variance and finite third absolute moments satisfies $\Delta \le C_{\mathrm{E}}\ell$ whenever $\ell \le \exp(-\exp(97))$, where Δ is the Kolmogorov distance of the standardized sum from the normal law, ℓ is its Lyapunov ratio, and $C_{\mathrm{E}} = (3+\sqrt{10})/(6\sqrt{2\pi})$. The coefficient is optimal. The proof develops the variational, uniform-jitter, and local-comparison methods of He and Cheng for summands with different laws. A finite approximate-period argument and a quantitative form of Esseen's moment inequality identify a common scale. Support variations put every coordinate of a maximizing array into two narrow clusters or a small neighborhood of zero. An exact comparison with a common-diameter Bernoulli array completes the proof. All parameters are explicit. The common-diameter bound and an auxiliary unrestricted bound are supplied by companion papers with validated finite calculations.

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1 The theorem and the proof

Let $X_1, \dots, X_n$ be independent centered real random variables. Put

$$B^2 = \sum_i \mathbb{E}X_i^2 > 0, \qquad T = \sum_i \mathbb{E}|X_i|^3 < \infty, \qquad \ell = T/B^3, \tag{1.1}$$

and write

$$\Delta = \sup_x \left| \mathbb{P}\left\{ B^{-1} \sum_i X_i \le x \right\} - \Phi(x) \right|, \qquad \mathcal{R}(X_1, \dots, X_n) = \Delta/\ell.$$

Here Φ and ϕ are the standard normal distribution function and density. Zero coordinates are allowed. Set $c_{\mathrm{E}} = 3 + \sqrt{10}$ and $C_{\mathrm{E}} = c_{\mathrm{E}}\phi(0)/6$.

Theorem 1.1. *For every array satisfying* (1.1)*,*

$$\boxed{\ell \le \ell_0 := \exp(-\exp(97)) \quad \Longrightarrow \quad \Delta \le C_{\mathrm{E}}\ell.}$$

The coefficient C_{E} cannot be decreased, even when all coordinates have the same two-point law.

A companion paper [2] classifies every full finite-horizon global maximizer below an ineffective Lyapunov threshold. That qualitative structural result concerns the maximizing laws; the present theorem gives an explicit value bound for every array.

The value of ℓ_0 is a sufficient threshold. The proof does not estimate the largest admissible threshold, and the sharp inequality for unrestricted independent arrays remains open.

He and Cheng [6, Theorem 1.1] prove the sharp inequality for iid sums once their sample size exceeds a universal threshold. We use the arguments in Sections 2–5 and Appendix A of their first version [5], and retain that version's numbering when citing its lemmas and equations. Their proof supplies three mechanisms used here: coordinate contamination gives support equations, uniform jitter reveals a lattice scale, and a local comparison separates small and large residual variance. Independence with different laws requires additional estimates. We prove these for the original coordinates, including those outside the two-cluster class. In particular, conditioning on a Bernoulli count does not make the residuals independent; our conditional Fourier argument retains each label together with its residual.

Two finite bounds from the companion papers enter the proof:

$$\mathcal{R}(h(B_1 - p_1), \dots, h(B_m - p_m)) \le C_{\mathrm{E}}, \tag{1.2}$$

$$\mathcal{R}(X_1, \dots, X_n) < 9/20. \tag{1.3}$$

In (1.2), the B_i are independent Bernoulli variables with arbitrary parameters and positive total variance, and $h > 0$. This is the common-diameter theorem [1, Theorem 1.2(ii)]; its proof uses Schulz's binomial theorem [8, Theorem 1] and validated finite calculations. Equation (1.3) follows from the unrestricted bound [3, Theorem 1.1]. These are the computational dependencies of the present theorem. We also use Shevtsova's general-independent estimate

$$\Delta \le C_{\mathrm{E}}\ell + 0.2538\ell^{4/3} \qquad (\ell \le 10^{-3}) \tag{1.4}$$

[9, Corollary 4.18, p. 303]. Its role is to localize a maximizing counterexample. Its positive remainder is removed by the subsequent support and local comparisons.

For the lattice step, define

$$p_{\mathrm{E}} = \frac{4 - \sqrt{10}}{2} = \frac{1 - 1/c_{\mathrm{E}}}{2}, \quad q_{\mathrm{E}} = 1 - p_{\mathrm{E}}, \quad \tau_{\mathrm{E}} = p_{\mathrm{E}}^2 + q_{\mathrm{E}}^2, \quad h_{\mathrm{E}} = 1/\tau_{\mathrm{E}}. \tag{1.5}$$

Esseen's moment inequality [4, pp. 162–163] states that a centered variable Y supported on a translate of $h\mathbb{Z}$ satisfies

$$|\mathbb{E}Y^3| + 3h\mathbb{E}Y^2 \le c_{\mathrm{E}}\mathbb{E}|Y|^3. \tag{1.6}$$

For positive variance, equality holds precisely for $Y \overset{d}{=} \pm h(B - p_{\mathrm{E}})$, $B \sim \mathrm{Bern}(p_{\mathrm{E}})$. Appendix B proves the quantitative form needed here. The iid asymptotic formula for this law gives a ratio tending to C_{E} and a Lyapunov ratio tending to zero [4, 7]. This proves optimality.

We use closed and strict CDFs $F_S(x) = \mathbb{P}(S \le x)$ and $F_S^-(x) = \mathbb{P}(S < x)$. For a centered sum of variance B^2,

$$\sup_x |F_S(x) - \Phi(x/B)| = \sup_x \max\{F_S(x) - \Phi(x/B),\ \Phi(x/B) - F_S^-(x)\}. \tag{1.7}$$

Reflection interchanges these branches. The comparisons below retain these endpoint conventions and therefore include laws with atoms.

Section 2 selects an attained counterexample with small Lyapunov ratio and bounds every support. Section 3 finds an approximate period by a finite procedure. Section 4 converts its moment information into restrictions on every support point. Section 5 proves the independent local comparison with explicit parameters, and Section 6 evaluates their composition. The appendices give the Fourier estimates, the quantitative support argument, and the conditional-moment estimates used in these sections.

2 Selection and support equations

Normalize total variance to one, so $\ell = T$. For a fixed number of coordinates, maximize the positive signed ratio at a threshold, less a nondecreasing penalty on ℓ.

Lemma 2.1 (Localized selection)**.** *Let $0 < \eta \le 1/100$ and $0 < t \le \min\{1/2000, \eta^3/108\}$. If a finite array has $\ell \le t$ and $\mathcal{R} > C_{\mathrm{E}}$, there is an attained penalized maximizer with the same number of coordinates, a threshold z, and*

$$C_{\mathrm{E}} < R < 9/20, \quad \ell < 2t, \quad R \le K := R + \ell\psi_t'(\ell) \le R + \eta. \tag{2.1}$$

Here R is its full ratio. If a coordinate has variance s and third absolute moment b, then

$$\psi_t\left(\frac{\ell - b}{(1-s)^{3/2}}\right) - \psi_t(\ell) \le 2\eta s. \tag{2.2}$$

Proof. Put $f(v) = 0$ for $v \le 0$, $f(v) = v^2/(1+v)$ for $v > 0$, and $\psi_t(u) = 2u^{1/3} f(\log(u/t))$. This penalty is continuously differentiable, nonnegative, nondecreasing, and zero for $u \le t$. After reflection if necessary, the supremum of

$$[F_S(z) - \Phi(z)]/\ell - \psi_t(\ell) \tag{2.3}$$

over centered variance-one arrays of the specified length is above C_{E}. If the objective exceeds $r_0 > 0$, then $n^{-1/2} \le \ell \le r_0^{-1}$. Bounded third moments give coordinatewise weak subsequences and uniform integrability of squares, preserving means and total variance. Chebyshev bounds the thresholds because their discrepancies are at least $r_0/\sqrt{n}$. Moving-threshold Portmanteau for the closed CDF gives the required upper semicontinuity at a limit. A strict loss of third moments would increase the positive ratio and decrease the penalty, contradicting the supremum. Thus the limit attains it. This is the fixed-dimension argument of He and Cheng [5, Lemma 4.1], applied coordinatewise.

Since $f(\log 2) > 4/15 > 1/4$, one has $\psi_t(u) > u^{1/3}/2$ for $u \ge 2t$. For $u \le 10^{-3}$ this exceeds the excess in (1.4); for $u \ge 10^{-3}$ it is at least $1/20 > 9/20 - C_{\mathrm{E}}$. Hence the maximizer has $\ell < 2t$. For $t < u \le 4t$, $f < 1$ and $0 \le f' \le 1$ give

$$0 \le u\psi_t'(u) \le \tfrac{8}{3}(4t)^{1/3} < 3(4t)^{1/3} \le \eta.$$

This holds also for $u \le t$. Each coordinate has $s^{3/2} \le \ell < 10^{-3}$, so $s \le 1/4$ and $\ell_- = (\ell - b)/(1-s)^{3/2} < (8/5)\ell < 4t$. If $\ell_- > \ell$, integration on this actual interval gives

$$\psi_t(\ell_-) - \psi_t(\ell) \le \eta \log(\ell_-/\ell) \le \tfrac{3}{2}\eta[-\log(1-s)] \le 2\eta s.$$

Otherwise monotonicity suffices. A larger discrepancy in the selected array, reflected for the other branch, would increase its objective; therefore R is its full ratio. □

Write $s_i = \mathbb{E}X_i^2$, $b_i = \mathbb{E}|X_i|^3$, $M_i = \mathbb{E}[X_i|X_i|]$, and let G_i be the CDF of $S - X_i$.

Lemma 2.2 (Support contact)**.** *For every $y \in \mathbb{R}$,*

$$\begin{aligned} D_i(y) := G_i(z-y) - F_S(z) + \phi(z)y + \frac{z\phi(z)}{2}(y^2 - s_i) \\ + \frac{3}{2}K\ell(y^2 - s_i) - K\{|y|^3 - b_i - 3M_i y\} \le 0. \end{aligned} \tag{2.4}$$

Equality holds at every $y \in \operatorname{supp} X_i$.

Proof. Replace the raw law of coordinate i by $(1-e)\operatorname{Law}(X_i) + e\delta_y$, retain the raw threshold, and center and standardize the new array. The derivatives at $e = 0$ of its mean, variance, and total third absolute central moment are y, $y^2 - s_i$, and $|y|^3 - b_i - 3M_i y$. The last differentiation is justified by a bound $C_y(1 + X_i^2)$. The raw CDF derivative is $G_i(z-y) - F_S(z)$. Differentiating (2.3) and multiplying by ℓ gives (2.4), with its sign supplied by maximality. Its integral under $\operatorname{Law}(X_i)$ is zero. Upper semicontinuity of D_i then extends equality to the entire support. This is the independent-coordinate form of the contamination argument of He and Cheng [5, Lemma 4.3]. □

Lemma 2.3 (Uniform support bound)**.** *Every nonzero coordinate of the selected maximizer satisfies*

$$b_i < 3\ell s_i, \qquad s_i < 9\ell^2, \qquad \operatorname{supp} X_i \subset (-8\ell, 8\ell). \tag{2.5}$$

Proof. Deletion, padded by a zero coordinate, stays in the maximization class. Its ratio at every threshold is at most $R + 2\eta s_i$. Compare the convolution of the deleted normal approximation with X_i to its convolution with $N(0, s_i)$. Put $a = \phi(0)(1 + 4\phi(0))/6 < 13/75$. Translation Taylor expansion and $\mathbb{E}|N(0, s_i)|^3 = 4\phi(0)s_i^{3/2} \le 4\phi(0)b_i$ give

$$R\ell(1 - s_i)^{3/2} \le (R + 2\eta s_i)(\ell - b_i) + ab_i.$$

Using $1 - (1 - s_i)^{3/2} \le 3s_i/2$ and rearranging gives

$$(R - a)b_i \le (3R/2 + 2\eta)\ell s_i.$$

Since $R > 2/5$, the right coefficient is less than $3(R - a)$. This proves $b_i < 3\ell s_i$, and Jensen gives $\sqrt{s_i} < 3\ell$.

For $0 \le s \le 1/4$ put

$$H_s(z, y) = \Phi\left(\frac{z - y}{\sqrt{1 - s}}\right) - \Phi(z) + \phi(z)y + \tfrac{1}{2}z\phi(z)(y^2 - s).$$

Translation and scale Taylor expansions give

$$|H_s(z, y)| \le \phi(0)|y|^3/6 + \phi(0)s|y|/2 + s^2/6, \tag{2.6}$$

$$|H_s(z, y) - \phi''(z)(sy/2 - y^3/6)| \le y^4/36 + sy^2/6 + s^2/6. \tag{2.7}$$

Use $\|\phi''\|_\infty = \phi(0)$, $\|\phi'''\|_\infty \le 2/\pi < 2/3$; the scale remainder is bounded in absolute value by $4s^2/27 < s^2/6$. Substitute the deleted bound in the contact equality. Its constant terms are

$$R(\ell_- - \ell) - \tfrac{3}{2}K\ell s + Kb + [\psi_t(\ell_-) - \psi_t(\ell)]\ell_-.$$

Since $(1 - s)^{-3/2} < 8/5$ and $(1 - s)^{-3/2} - 1 - 3s/2 \le 6s^2$, every support point satisfies

$$0 \le H_s(z, y) - K|y|^3 + \tfrac{3}{2}K\ell y^2 + 3KM_i y + 6R\ell s^2 + \eta b + 4\eta\ell s. \tag{2.8}$$

The contact equation first bounds the support, since its negative cubic term dominates at infinity. Let A_i be its maximum absolute value. Use $|M_i| \le s$, $b \le sA_i$, $K \le 23/50$, and (2.6) to obtain

$$A_i^3/3 \le (7/10)\ell A_i^2 + (8/5)sA_i + (3/5)s^2 + (1/25)\ell s.$$

With $s \le 9\ell^2$, $\ell \le 1/8$, and $x = A_i/\ell$,

$$x^3/3 \le 7x^2/10 + 72x/5 + 1287/200.$$

The difference is $2539/600 > 0$ at $x = 8$ and has positive derivative for $x \ge 8$. Thus $A_i < 8\ell$. □

3 An approximate period and uniform jitter

A uniform random shift suppresses the nonzero lattice peaks of a characteristic function. Its period must be chosen from the array itself. The following finite construction gives an explicit error.

Lemma 3.1 (Finite period selection)**.** *Let μ be a probability measure on $[-M, M]$, $M > 0$, and put*

$$g(u) = \int \frac{|e^{iud} - 1|^2}{d^2}\,\mu(dd), \quad q(u) = \sqrt{g(u)}, \quad \delta = 1/(4M),$$

where the integrand at zero is u^2. For $T \geq \delta$ and $0 < r \leq \delta/4$ define

$$B_0 = T/\delta + 2, \quad J = \lceil \log_2(T/\delta) \rceil + 1, \quad \varepsilon = r/(2B_0^J), \quad E = \{u \in [\delta, T] : q(u) \leq \varepsilon\}.$$

If $E \neq \varnothing$, there is $p \in [\delta, T]$ such that $q(p) \leq r/2$ and $\operatorname{dist}(u, p\mathbb{Z}) < r$ for every $u \in E$. For $h = 2\pi/p$ and $\ell > 0$,

$$\int_\delta^T e^{-g(u)/(2\ell^2)} |\operatorname{sinc}(hu/2)|\,\frac{du}{u} \leq \frac{56\sqrt{\pi}}{9}\frac{r\ell}{p^2} + \log(T/\delta) e^{-\varepsilon^2/(2\ell^2)}. \tag{3.1}$$

If $E = \varnothing$, take $h = 0$; the exponential term alone suffices.

Proof. The Hilbert-space triangle inequality gives subadditivity of q, $q(-u) = q(u)$, and $q(u) \leq |u|$. Thus q is 1-Lipschitz. Also $g'' \geq 2 - M^2 g$ and $g(u) \geq u^2/2$ for $|u| \leq \delta$. Start with $p_0 \in E$. If $v \in E$ has distance at least r from $p_j\mathbb{Z}$, choose the nearest integer k and put $p_{j+1} = |v - kp_j|$. While $p_j \geq \delta$,

$$r \leq p_{j+1} \leq p_j/2, \qquad q(p_{j+1}) \leq \varepsilon + (T/\delta + 1/2) q(p_j) \leq B_0^{j+1}\varepsilon.$$

A proposed value in $[r, \delta]$ would satisfy both $q \leq r/2$ and $q \geq r/\sqrt{2}$. There are at most $\lceil \log_2(T/\delta) \rceil$ halvings before this happens. The construction therefore stops with the required p.

Retain each $I_k = [kp - r, kp + r] \cap [\delta, T]$, $k \geq 1$, that meets E. On it $q \leq \varepsilon + 2r \leq 5\delta/8$, so $g'' \geq 1$. At its minimum a_k, including an endpoint minimum, $g(u) \geq (u - a_k)^2/2$. Moreover $|\operatorname{sinc}(\pi u/p)| \leq r/u$ and $u \geq (k - 1/4)p$. Its integral is at most $2\sqrt{\pi}\ell r/[(k - 1/4)^2 p^2]$. Sum using $\sum_{k \geq 1}(k - 1/4)^{-2} \leq 28/9$. Outside the retained intervals $g > \varepsilon^2$, proving the assertion. □

Proposition 3.2 (Uniform jitter)**.** *Suppose $\sum \mathbb{E}X_i^2 = 1$, $\ell = \sum \mathbb{E}|X_i|^3 \leq 1/8$, and $|X_i| \leq 8\ell$. For $0 < \rho \leq 1$ put*

$$\begin{aligned} \delta &= 1/64, & T &= 80/\rho, & r &= \rho\delta^2/48, \\ B_0 &= T/\delta + 2, & J &= \lceil \log_2(T/\delta) \rceil + 1, & \varepsilon &= r/(2B_0^J), \\ L_{\mathrm{jit}}(\rho) &= \min\left\{\frac{1}{8}, \frac{\rho}{48000}, \frac{\rho\varepsilon^2}{8\log(T/\delta)}\right\}. \end{aligned} \tag{3.2}$$

If $\ell \leq L_{\mathrm{jit}}(\rho)$, there is $0 \leq h \leq 128\pi$ such that, for an independent U_h uniform on $[-h/2, h/2]$,

$$\left\| F_{S + \ell U_h}(x) - \Phi(x) - \frac{\kappa}{6}(1 - x^2)\phi(x) \right\|_\infty \leq \rho\ell, \qquad \kappa = \sum_i \mathbb{E}X_i^3. \tag{3.3}$$

When $h > 0$, $g(2\pi/h) \leq r^2/4$ for the measure below. Here $U_0 = 0$.

Proof. Set $Y_i = X_i/\ell$, with independent copies Y_i'. The measure

$$\mu(A) = \frac{\ell^2}{2}\sum_i \mathbb{E}[(Y_i - Y_i')^2 \mathbf{1}_{\{Y_i - Y_i' \in A\}}] \tag{3.4}$$

is a probability measure on $[-16, 16]$. It satisfies

$$g(u) = \ell^2 \sum_i (1 - |\mathbb{E}e^{iuY_i}|^2), \qquad |\mathbb{E}e^{iuS/\ell}| \le e^{-g(u)/(2\ell^2)}.$$

Apply Lemma 3.1. The low-frequency and signed-smoothing estimates in Appendix A bound the norm in (3.3) by

$$12000\ell^2 + 12r\ell/\delta^2 + \log(T/\delta)e^{-\varepsilon^2/(2\ell^2)} + 20\ell/T. \tag{3.5}$$

Each term is at most $\rho\ell/4$ under (3.2); for the third use $e^{-x} \le 1/x$. □

4 From moment stability to complete supports

The contact equations strengthen the moment estimates into bounds for each coordinate's full support.

Proposition 4.1 (Two-cluster reduction)**.** *Let $0 < \lambda \le 1$ and $N \ge 1$. Set*

$$\rho = 10^{-26}\lambda^2, \qquad \eta = 10^{-10}\lambda, \qquad L_* = \min\{L_{\mathrm{jit}}(\rho), 1/(2\sqrt{N})\}. \tag{4.1}$$

An attained maximizer of Lemma 2.1 with this penalty bound and $\ell \le L_*$ can be represented, after division by ℓ, as independent coordinates

$$Y_i = h_i(B_i - p_i) + U_i \quad (1 \le i \le m), \qquad Z_1, \dots, Z_s, \tag{4.2}$$

where $m \ge N$, $h > 0$, $B_i \sim \mathrm{Bern}(p_i)$, and

$$p_i \in [1/4, 3/4], \quad |h_i/h - 1| \le \lambda, \quad \mathbb{E}(U_i \mid B_i) = 0, \quad |U_i| \le \lambda h, \tag{4.3}$$

$$\mathbb{E}Z_j = 0, \quad |Z_j| \le \lambda h, \quad \sum_j \mathrm{Var}\, Z_j \le \lambda m h^2. \tag{4.4}$$

The representations are coordinatewise. Each pair (B_i, U_i) is independent of the other pairs and of all Z_j.

The proof is in Appendix B. Rounding at the period from Proposition 3.2 loses a controlled total second moment. The deficit in Esseen's inequality forces that period close to h_{E} and the maximizing threshold close to zero. Using the same jitter for every deletion shows that its one-sided smoothing loss is small at each support point. Comparing two such losses restricts every difference of support points to a small neighborhood of $h\mathbb{Z}$. The cubic expansion of (2.8) bounds each scaled support diameter by $7/2$ and puts the two-cluster probabilities in $[1/4, 3/4]$. Coordinates with small diameter have small total variance by moment stability. No lower bound on individual atom masses enters the argument.

5 An exact comparison near a common diameter

Theorem 5.1 (Explicit independent local comparison)**.** *Set*

$$\mathcal{A} = 38e^{43}, \qquad \lambda = e^{-\mathcal{A}}, \qquad N = \lceil e^{\mathcal{A}} \rceil. \tag{5.1}$$

Every independent array of the form (4.2)–(4.4), with $m \ge N$, has $\mathcal{R} \le C_{\mathrm{E}}$. Its ratio uses the total variance and the sum of third absolute moments of all the original coordinates.

We follow the two residual regimes of He and Cheng [5, Section 3]. The estimates below are uniform in unequal probabilities, unequal gaps, and the additional coordinates.

5.1 Normalization and conditional estimates

Put $v_i = p_i(1-p_i)$ and define the variance-weighted mean gap

$$\bar{h} = \frac{\sum_i v_i h_i}{\sum_i v_i}.$$

Then $(1-\lambda)h \le \bar{h} \le (1+\lambda)h$. Divide every coordinate by $\bar{h}$, retaining h_i, U_i, Z_j for the rescaled gaps and residuals. Write

$$V = \sum_i v_i, \quad \mu = \sum_i p_i, \quad h_i = 1 + a_i, \quad \sum_i v_i a_i = 0,$$
$$K_0 = \sum_i B_i, \quad A = \sum_i a_i(B_i - p_i), \quad U = \sum_i U_i + \sum_j Z_j, \quad W = A + U, \quad S = K_0 + W,$$
$$\Lambda_A = \sum_i v_i a_i^2, \quad \Lambda_U = \sum_i \mathbb{E}U_i^2 + \sum_j \mathbb{E}Z_j^2, \quad \Lambda = \Lambda_A + \Lambda_U, \quad B^2 = V + \Lambda.$$

The uncentered sum S has mean μ, and $\mathrm{Cov}(K_0, W) = 0$. Put $\varepsilon_0 = \max(\max_i \|U_i\|_\infty, \max_j \|Z_j\|_\infty)$, with an empty maximum zero, and $\xi = \Lambda/V$. The hypotheses imply

$$3m/16 \le V \le m/4, \quad \max_i |a_i| \le 3\lambda, \quad \varepsilon_0 \le 2\lambda, \quad \xi \le 8\lambda, \quad \Lambda \le 2\lambda m. \tag{5.2}$$

For example the upper bound before enlargement is $\xi \le [\lambda + (7/4)\lambda^2]/[(3/16)(1-\lambda)^2]$.

Let T_X and κ be the total absolute and signed third central moments of the original array. Its reference Bernoulli moments are

$$T_0 = \sum_i v_i(p_i^2 + (1-p_i)^2), \qquad \kappa_0 = \sum_i v_i(1-2p_i).$$

Conditional centering gives

$$|T_X - T_0|,\ |\kappa - \kappa_0| \le 2\sqrt{m\Lambda_A} + 4\Lambda, \tag{5.3}$$
$$T_X \ge V/3, \quad 1/4 \le T_X/B^2 \le 2, \quad B^2 \le m, \quad B/2 \le P := B^3/T_X \le 4B. \tag{5.4}$$

Indeed $|(1+a)^3 - 1| \le 4|a|$ and $\sum v_i|a_i| \le \sqrt{V\Lambda_A}$. Within each cluster the sign is fixed: its unperturbed magnitude is at least $7/32 > \varepsilon_0$. Expanding the cubic changes either third moment by at most $4\mathbb{E}U_i^2$, since the linear term vanishes. An extra coordinate contributes at most $\varepsilon_0 \mathbb{E}Z_j^2$. Jensen gives $T_X \ge (7/8)^3 T_0 \ge 343V/1024 > V/3$. Every original coordinate has absolute value at most two, so $T_X \le 2B^2$, proving (5.4).

For $m \ge 10^{12}$ and $|k - \mu| \le 6\sqrt{m}$, Appendix C proves

$$|m_k| := |\mathbb{E}(A \mid K_0 = k)| \le 10^4 \sqrt{\Lambda_A}/\sqrt{m}, \tag{5.5}$$
$$|\operatorname{Var}(A \mid K_0 = k) - \Lambda_A| \le 40000\Lambda_A/\sqrt{m},$$
$$\mathbb{E}[(A - m_k)^4 \mid K_0 = k] \le 10^7 \Lambda_A^2. \tag{5.6}$$

It also proves the relative label bound

$$\left| \frac{\mathbb{P}(B_i = b \mid K_0 = k)}{\mathbb{P}(B_i = b)} - 1 \right| \le 500/\sqrt{m} \qquad (b = 0, 1). \tag{5.7}$$

For $\Lambda > 0$ define

$$D = \Lambda^2 + \varepsilon_0^2 \Lambda_U, \qquad H = \max\left\{ \sqrt{\Lambda_A}, \frac{\Lambda}{\sqrt{\Lambda + \varepsilon_0^2}} \right\}. \tag{5.8}$$

Then, in the same count range,

$$\begin{gathered}\mathbb{E}(W \mid K_0 = k) = m_k, \quad \Lambda/2 \le \operatorname{Var}(W \mid K_0 = k) \le 2\Lambda, \\ \mathbb{E}[(W - m_k)^4 \mid K_0 = k] \le 10^8 D, \quad \mathbb{E}(|W - m_k| \mid K_0 = k) \ge H/30000, \\ \mathbb{E}(W^4 \mid K_0 = k) \le 10^9 D, \qquad \mathbb{E}W^4 \le 400D.\end{gathered} \tag{5.9}$$

To see the extension, condition first on all labels. The noise then has variance $T_{\mathbf{b}} \le 4\Lambda_U$ and fourth moment at most $3T_{\mathbf{b}}^2 + \varepsilon_0^2 T_{\mathbf{b}}$. Its mean given K_0 is zero, and (5.7) compares its variance to Λ_U. Combine these facts with (5.6) and $|x+y|^4 \le 8(|x|^4+|y|^4)$. Interpolation $\mathbb{E}|Y| \ge (\mathbb{E}Y^2)^{3/2}/(\mathbb{E}Y^4)^{1/2}$ supplies the second term in H. Conditional Jensen given the labels supplies the first from $A - m_k$. In particular $|m_k| \le 10^4 H/\sqrt{m}$ uniformly, even when Λ_A is much smaller than ε_0^2/m. Finally $\max a_i^2 \le 16\Lambda_A/3$ gives $\mathbb{E}A^4 \le 9\Lambda_A^2$, and hence the unconditional bound.

5.2 Uniform approximation and central thresholds

Let J be the CDF of S plus an independent uniform variable on $[-1/2, 1/2]$. If $T \ge 10$ and $\xi(T+1)^2 \le 1/56$, put

$$\begin{aligned}E_m(T) = \frac{401(1+\log T)}{\sqrt{m}} &+ (1+\log T)\xi(T+1)^2 \\ &+ 60000(1+\log T)\sqrt{m}\, e^{-m/262144} + 56/T.\end{aligned} \tag{5.10}$$

Appendix A proves, for $z = (x-\mu)/B$,

$$\sqrt{m}\left\| J(x) - \Phi(z) - \frac{\kappa}{6B^3}(1-z^2)\phi(z)\right\|_\infty \le E_m(T). \tag{5.11}$$

The expansion includes the third moments of the additional coordinates.

Fix

$$a_0 = e^{43}, \quad \mathcal{C} = \frac{49}{4}a_0, \quad \mathcal{B} = \frac{25}{2}a_0, \quad q = e^{-\mathcal{C}}, \quad T = e^{\mathcal{B}}.$$

We check the four error terms in (5.10) directly. For every $m \ge N$,

$$\xi(T+1)^2 < 1/56, \quad E_m(T) < q/1000, \quad m^{-1/2} < q/1000, \quad 200\lambda < q/1000. \tag{5.12}$$

For a direct check, the four terms in (5.10) are at most

$$802\mathcal{B}e^{-\mathcal{A}/2}, \quad 64\mathcal{B}e^{-\mathcal{A}+2\mathcal{B}}, \quad 240000\mathcal{B}(262144)^2 e^{-3\mathcal{A}/2}, \quad 56e^{-\mathcal{B}}.$$

The third uses $e^{-x} \le 2/x^2$. Dividing these bounds by $q/4000$ gives, respectively,

$$3208000\mathcal{B}e^{-27a_0/4}, \quad 256000\mathcal{B}e^{-3a_0/4}, \quad 960000000\mathcal{B}(262144)^2 e^{-179a_0/4}, \quad 224000e^{-a_0/4}.$$

Each is less than one. The logarithms of the first three prefactors are below 110, since $\log\mathcal{B} = 43 + \log(25/2)$, whereas their smallest negative exponent has magnitude $3a_0/4 > 750$. The fourth follows from $a_0/4 > 250 > \log 224000$. The remaining inequalities in (5.12) follow from

$$\xi(T+1)^2 \le 64e^{-13a_0} < 1/56, \quad m^{-1/2}/q \le e^{-27a_0/4} < 1/1000, \quad 200\lambda/q = 200e^{-103a_0/4} < 1/1000.$$

These estimates hold without an upper bound on m.

Shift the jitter by $\pm 1/2$. Either normalized signed discrepancy is at most

$$\phi(z)\left\{\frac{B^2}{2T_X} + \frac{|\kappa|}{6T_X}|1-z^2|\right\} + 4E_m(T) + m^{-1/2}. \tag{5.13}$$

The normalized Taylor error is at most $1/(4B) < m^{-1/2}$, using $\|\phi'\|_\infty < 1/4$, $\|((1-z^2)\phi(z))'\|_\infty < 5/4$, and (5.4). The main term is at most $(3+z^2)\phi(z)$, below $1/100$ for $|z| \ge 4$. Thus the full bound there is below $1/24$.

5.3 Small residual variance

Suppose $0 < \Lambda \le e^{-43}$. If $\Lambda = 0$, the result is exactly (1.2). We use a one-sided loss estimate: if $\mathbb{E}Y = 0$, $\mathbb{E}Y^4 \le 1/160000$, $f(0) = 0$, and $3/4 \le f' \le 5/4$ on $[-7/10, 7/10]$, then for $|u| \le 3/5$,

$$\begin{aligned} \mathbb{P}(Y > u) + f(u) &\ge \tfrac{3}{8}\mathbb{E}|Y| - 176000\mathbb{E}Y^4, \\ \mathbb{P}(Y < u) - f(u) &\ge \tfrac{3}{8}\mathbb{E}|Y| - 176000\mathbb{E}Y^4. \end{aligned} \tag{5.14}$$

This is the clipping argument of He and Cheng [5, Lemma 3.2 and equation (83)]. Appendix C gives its proof with these constants and the stated threshold range.

The count estimate in Appendix A gives

$$b_k := \mathbb{P}(K_0 = k) \ge e^{-100}/\sqrt{m} \qquad (|k - \mu| \le 6\sqrt{m}). \tag{5.15}$$

Write a threshold with $|z| < 4$ as $k+u$, $|u| \le 1/2$, where k is a nearest integer. Then $|k-\mu| \le 5\sqrt{m}$. For $f_k(u) = [\Phi((k-\mu+u)/B) - \Phi((k-\mu)/B)]/b_k$, we have $3/4 \le f_k' \le 5/4$ on $[-1, 1]$. Indeed the relative count error is at most $18e^{100}/\sqrt{m}$, and the logarithm of the Gaussian density ratio is at most $100\xi + 30/\sqrt{m}$; both are below $1/100$.

The exact count decompositions are

$$\begin{aligned} F_S(k+u) &= F_{K_0}(k) - b_k\mathbb{P}(W > u \mid K_0 = k) + e^{+}_{k,u}, \\ F_S^{-}(k+u) &= F_{K_0}(k-1) + b_k\mathbb{P}(W < u \mid K_0 = k) + e^{-}_{k,u}. \end{aligned} \tag{5.16}$$

Each remainder is bounded in absolute value by

$$\sum_{j \ne k} b_j \mathbb{P}\{|W| \ge |j-k| - 1/2 \mid K_0 = j\} \le 2 \cdot 10^{11} D/\sqrt{m}.$$

For $|j-k| \le \sqrt{m}$, use $b_j \le 3/\sqrt{m}$, (5.9), and $\sum_{r \ne 0}(|r| - 1/2)^{-4} < 40$. For more distant counts use $\mathbb{E}W^4 \le 400D$ and a denominator at least $m^2/16$.

Apply (5.14) to $Y = W - m_k$ conditionally on $K_0 = k$, using $f_k(v + m_k) - f_k(m_k)$ and $v = u - m_k$. The bounds above give $|m_k| < 1/20$ and the required fourth-moment bound, since $10^8 D \le 2 \cdot 10^8 e^{-86} < 1/160000$. At a central threshold, $B > 50$ gives $|(k-\mu)/B| < 4.01$. Using $f_k'(0) \le 5/4$ and $P/B \ge 1/2$,

$$Pb_k \ge \frac{4}{5}\frac{P}{B}\phi((k-\mu)/B) \ge \frac{2}{5}\phi(4.01) > e^{-11}.$$

For the last inequality, $\sqrt{2\pi} < 3$, $4.01^2/2 < 17/2$, and $e^{5/2} > (8/3)^2(3/2) > 15/2$ suffice. The positive term in (5.14) is at least $H/80000$, so its normalized loss exceeds $e^{-11}H/80000 > e^{-23}H$; the last comparison follows from $(8/3)^{12} > 80000$. Also $Pb_k \le 5$. Its fourth-moment and leakage errors are below $10^{14}D$, and its conditional-mean error is at most $62500\sqrt{\Lambda_A}/\sqrt{m}$.

Let $R_0 = \mathcal{R}(B_1 - p_1, \dots, B_m - p_m) \le C_{\mathrm{E}}$. Equation (5.3) gives

$$\left| \frac{B^3/T_X}{V^{3/2}/T_0} - 1 \right| \le 32\sqrt{\Lambda_A}/\sqrt{m} + 96\Lambda/m.$$

The normal variance change costs at most $\xi/8$ before multiplication by P. Combining these errors with (5.16), either central normalized branch is at most

$$R_0 - e^{-23}H + e^{40}\left\{ D + \frac{\sqrt{\Lambda_A} + \Lambda}{\sqrt{m}} \right\}. \tag{5.17}$$

The full definition of H gives

$$D/H \le (\Lambda + \varepsilon_0^2)^{3/2}, \quad \sqrt{\Lambda_A}/H \le 1, \quad \Lambda/H \le \sqrt{\Lambda + \varepsilon_0^2}.$$

Since $\varepsilon_0 \le e^{-43}$ and $m \ge e^{1000}$, the error is at most $e^{40}(3e^{-129/2} + 2e^{-500})H$. Its ratio to $e^{-23}H$ is at most

$$3e^{-3/2} + 2e^{-437} < 3/4 + 2^{-436} < 1,$$

using $e > 8/3$ and $\sqrt{e} > 3/2$. Thus the central comparison is strict for every positive residual variance. Finally $R_0 \ge \phi(0)/4 > 1/12$: apply the count estimate at an integer nearest μ and take half its jump. The bound $1/24$ at distant thresholds is also below R_0.

5.4 Positive residual variance and interval mass

Suppose $\Lambda \ge e^{-43}$ and $|z| \le 4$. Choose all integers $|k - x| \le \max(1, \sqrt{\Lambda})$. There are at least $\sqrt{\Lambda}$ of them, all within $6\sqrt{m}$ of μ. Tilt the Bernoulli probabilities exponentially to make their count mean k, and project the gap residual off the count, as in Appendix C. On $K_0 = k$ the residual is $m_k^0 + R_k$, where under the tilted measure R_k is a sum of independent centered pair residuals and additional coordinates, with

$$L := \mathrm{Var}_\theta\, R_k \in [\Lambda/2, 2\Lambda], \quad \mathrm{Var}_\theta\, K_0 \ge m/10, \quad |m_k^0| \le \sqrt{\Lambda}, \quad \mathrm{Cov}_\theta(K_0, R_k) = 0. \tag{5.18}$$

Each residual summand is bounded by 8λ. The within-cluster noise variances change relatively by at most $120/\sqrt{m}$ under the tilt; the extra coordinates do not change. Thus these assertions include their actual variances.

Apply Lemmas A.2 and A.3 to both quarter-intervals $(1/4, 1/2)$ and $(-1/2, -1/4)$, with

$$L_0 = \tfrac{1}{2}e^{-43}, \quad C_0 = 4 + 2e^{43/2}, \quad A_g = 4e^{43/2}, \quad \omega = 16e^{A_g^2/6}, \quad T_g = 4\omega.$$

The shift $y = x - k - m_k^0$ satisfies $|y| \le C_0\sqrt{L}$, and $A_g \ge C_0 + (7/16)/\sqrt{L_0}$. The interval minorant has Gaussian expectation at least $\phi(A_g)/(4\omega\sqrt{L})$. With $\beta = 7/64$, the three terms in (A.5) are at most

$$128e^{-19a_0}, \qquad 128e^{43/2 - 38a_0}, \qquad 10^7 e^{-34a_0}.$$

Their sum is below $e^{-10a_0} < \phi(A_g)/32$, because $\phi(A_g) > e^{-9a_0}$, $\omega < e^{3a_0}$, and $T_g < e^{4a_0}$. Indeed, after division by e^{-10a_0}, each term is below $1/3$, using $a_0 > 1000$ and $\log a_0 = 43$. Also $8\lambda T_g < 8e^{-34a_0} < \beta/162$. For the third term use $L \le m$, $e^{-x} \le 2/x^2$, and $m \ge N$; this is uniform over all possible residual variances.

Each conditional quarter-interval probability is at least $\phi(A_g)/(8\omega\sqrt{L})$. Multiply by the count probabilities (5.15) and sum over the chosen integers. Since $L \le 2\Lambda$, the resulting lower bound is

$$\frac{e^{-100}\phi(A_g)}{8\sqrt{2}\,\omega\sqrt{m}} > \frac{e^{-12a_0 - 104}}{\sqrt{m}} > \frac{q}{\sqrt{m}},$$

where the last inequality uses $\mathcal{C} - 12a_0 = a_0/4 > 104$. Thus

$$\mathbb{P}\{S - x \in (1/4, 1/2)\}, \quad \mathbb{P}\{S - x \in (-1/2, -1/4)\} \ge q/\sqrt{m}. \tag{5.19}$$

The two actual one-sided jitter losses therefore satisfy

$$J(x + 1/2) - F_S(x) \ge q/(2\sqrt{m}), \qquad F_S^-(x) - J(x - 1/2) \ge q/(2\sqrt{m}).$$

After multiplication by $P \geq B/2$, each loss exceeds $q/10$. Esseen's Bernoulli inequality gives $3V + |\kappa_0| \leq c_{\mathrm{E}} T_0$. Using (5.3), the main term of (5.13) is consequently at most

$$C_{\mathrm{E}} + 20\sqrt{\Lambda_A/m} + 42\Lambda/m \leq C_{\mathrm{E}} + 200\lambda.$$

Either central branch is below $C_{\mathrm{E}} - q/10 + 200\lambda + 4E_m(T) + m^{-1/2} < C_{\mathrm{E}} - q/20$. The distant thresholds were already bounded by $1/24$. This completes the proof of Theorem 5.1.

6 Evaluation of the threshold

Take λ, N from (5.1) and ρ, η, L_* from (4.1). A violation with Lyapunov ratio at most

$$t_* := \min\{L_*/2, \eta^3/108, 1/2000\} \tag{6.1}$$

would give, by Lemma 2.1, an attained violating maximizer with $\ell < 2t_* \leq L_*$. Proposition 4.1 puts the entire array into the class of Theorem 5.1, contradicting $R > C_{\mathrm{E}}$.

It remains to show $\ell_0 < t_*$. Write $A = \mathcal{A} = 38e^{43}$, so $\lambda = e^{-A}$, $N = \lceil e^A \rceil$, $\rho = 10^{-26}e^{-2A}$, and $\eta = 10^{-10}e^{-A}$. For the jitter parameters put

$$Q = T/\delta = 5120/\rho, \quad B_0 = Q + 2, \quad J = \lceil \log_2 Q \rceil + 1, \quad r = \rho/(48 \cdot 4096), \quad \varepsilon = r/(2B_0^J).$$

Since $A > 1000$, $\log 2 > 2/3$, and $\log 10 < 3$,

$$\log(1/\rho),\ \log Q,\ \log B_0,\ \log(1/r) < 3A, \qquad J < 5A.$$

The ceiling is included: $J < \log Q/\log 2 + 2 < 5A$. Thus $\log(1/\varepsilon) < 3A + 1 + 15A^2 < 16A^2$. The negative logarithm of the third entry defining L_{jit} is

$$\log(1/\rho) + 2\log(1/\varepsilon) + \log(8\log Q) < 32A^2 + 4A < 40A^2.$$

The first two entries have smaller negative logarithms. Also $N < 2e^A$, so $\log(2\sqrt{N}) < A$ and $-\log L_* < 40A^2$. The other entries of (6.1) have negative logarithms

$$3A + 30\log 10 + \log 108 < 4A, \qquad \log 2000 < A.$$

Therefore $-\log t_* < 41A^2 = 59204e^{86}$. The positive exponential series gives $e > 163/60$, and the rational comparison $(163/60)^{11} > 59204$ gives

$$-\log \ell_0 = e^{97} > 59204e^{86} > -\log t_*.$$

This completes the proof of Theorem 1.1.

A Fourier estimates with numerical constants

We use $\widehat{\nu}(t) = \int e^{itx}\nu(dx)$ and $\operatorname{sinc} u = \sin(u)/u$, with its continuous value at zero. This appendix supplies the quantitative versions of the signed smoothing and jitter arguments used in Sections 3 and 5.

A.1 Signed smoothing and the low-frequency expansion

If F is a probability CDF with finite first absolute moment and G is the CDF primitive of a finite signed measure of mass one and finite first absolute moment, with Lipschitz constant M_G, then

$$\|F - G\|_\infty \le \frac{1}{4}\int_{-A}^{A} \frac{|\widehat{F}(t) - \widehat{G}(t)|}{|t|}\, dt + 24 M_G/A. \tag{A.1}$$

Here $\widehat{F}$ and $\widehat{G}$ denote the transforms of their measures. This is the signed smoothing lemma of He and Cheng [5, Lemma 2.1 and equation (71)]. Its proof uses the even density $k(x) = 3\,\mathrm{sinc}(x/4)^4/(8\pi)$, whose transform is supported on $[-1,1]$. The bound $k(x) \le 3\min(1, 256/x^4)/(8\pi)$ gives $\mathbb{E}|Z| \le 12/\pi < 4$. With shift $a = 16$, its one-sided tail ε_k is below $1/8$, and $C_a = \mathbb{E}(16 - Z)_+ < 18$. At a positive near-maximum of $F - G$, monotonicity of F gives $(F-G)(x+y) \ge (F-G)(x) - M_G y$ for $y \ge 0$. Average with the shifted kernel; use $-\|F-G\|_\infty$ on its opposite tail. The negative branch uses the reflected shift. This bounds the norm by

$$\frac{\|(F-G) * k_A\|_\infty}{1 - 2\varepsilon_k} + \frac{C_a M_G}{A(1 - 2\varepsilon_k)}.$$

Fourier inversion of the integrable CDF difference gives coefficients $1/[2\pi(1-2\varepsilon_k)] \le 1/4$ and $C_a/(1-2\varepsilon_k) \le 24$. Only F needed monotonicity. In particular the weaker coefficients $1/\pi$ and 40 are valid.

Consider the variance-one row of Proposition 3.2. Write $v_i = \mathbb{E}X_i^2$, $\kappa = \sum \mathbb{E}X_i^3$, and $f_i(t) = \mathbb{E}e^{itX_i}$. For $|t| \le 1/(64\ell)$,

$$|f_i(t) - 1| \le v_i t^2/2 \le 1/128, \quad \sum_i \mathbb{E}X_i^4 \le 8\ell^2, \quad \sum_i v_i^2 \le 64\ell^2.$$

Using $|\log(1+z) - z| \le |z|^2$ for $|z| \le 1/2$ yields

$$\sum_i \log f_i(t) = -t^2/2 + \kappa(it)^3/6 + w(t), \qquad |w(t)| \le (49/3)\ell^2 t^4 \le t^2/6.$$

Thus the error against $\widehat{\nu}(t) = e^{-t^2/2}(1 + \kappa(it)^3/6)$ is at most

$$\ell^2 e^{-t^2/3}\{(49/3)t^4 + t^6/72\}. \tag{A.2}$$

Multiplication by $\mathrm{sinc}(h\ell t/2)$ adds at most $(h^2\ell^2/24)(t^2 + \ell|t|^5/6)e^{-t^2/2}$. The positive-frequency integrals after division by t are at most $(591/8)\ell^2$ and $(13h^2/288)\ell^2$, respectively. For $h \le 128\pi$ their sum is below $7500\ell^2$. The Gaussian comparison tail starting at $a = 1/(64\ell)$ is at most

$$\ell^2\{64^2 + 64/3\} < 4118\ell^2.$$

Here use $\int_a^\infty e^{-t^2/2}dt/t \le a^{-2}$ and $\int_a^\infty t^2 e^{-t^2/2}dt \le 2/a$. The comparison CDF $\Phi(x) + \kappa(1 - x^2)\phi(x)/6$ has Lipschitz constant below $1/2$, since $\|\phi'''\|_\infty \le 2/\pi$ and $|\kappa| \le 1/8$. Applying the weaker smoothing coefficients $1/\pi, 40$, and Lemma 3.1, now proves (3.5).

For the local calculation a smaller low-frequency bound is convenient. Put $m_{4,i} = \mathbb{E}X_i^4$ and $A_i = -v_i t^2/2 + \mathbb{E}X_i^3(it)^3/6$. On the same interval, $|A_i| \le (25/48)v_i t^2 < 1/64$ and

$$|f_i(t) - e^{A_i}| \le m_{4,i} t^4/2.$$

Symmetrization gives $|f_i(t)| \le e^{-v_i t^2/3}$. Telescope the products; omitting an individual decay factor costs less than two. Since $\sum_i m_{4,i} \le 8\ell^2$,

$$|\widehat{F}(t) - \widehat{\nu}(t)| \le 8\ell^2 t^4 e^{-t^2/3} + (\ell^2 t^6/72) e^{-t^2/2}.$$

Integrating over both signs gives less than $73\ell^2$. Using $|\operatorname{sinc}(h\ell t/2) - 1| \le h^2\ell^2 t^2/24$ and the actual product bound adds at most $h^2\ell^2/8$. Hence

$$\int_{|t|\le 1/(64\ell)} \frac{|\operatorname{sinc}(h\ell t/2)\widehat{F}(t) - \widehat{\nu}(t)|}{|t|}\, dt \le (73 + h^2/8)\ell^2. \tag{A.3}$$

A.2 The count estimate

Lemma A.1. *For a sum K of independent Bernoulli variables, put $\mu = \mathbb{E}K$ and $V = \operatorname{Var} K$. If $V \ge 1/2$, then*

$$\left| \mathbb{P}(K = k) - \frac{\phi((k-\mu)/\sqrt{V})}{\sqrt{V}} \right| \le 6/V, \qquad \sup_k \mathbb{P}(K = k) < 1/\sqrt{V}. \tag{A.4}$$

If the probabilities lie in $[\alpha, 1-\alpha]$, let $\beta = \alpha(1-\alpha)$ and $\gamma = \phi(M/\sqrt{\beta})$. For $m \ge 144/(\beta\gamma^2)$ and $|k - \mu| \le M\sqrt{m}$, $\mathbb{P}(K = k) \ge \gamma/\sqrt{m}$.

Proof. For $Z_i = B_i - p_i$ and $v_i = p_i(1-p_i)$, on $[-\pi, \pi]$,

$$|\mathbb{E}e^{itZ_i}| \le e^{-2v_i \sin^2(t/2)} \le e^{-2v_i t^2/\pi^2}, \qquad |\mathbb{E}e^{itZ_i} - e^{-v_i t^2/2}| \le v_i(|t|^3/6 + t^4/32).$$

Telescope, retaining the decay from $V - v_i \ge V/2$. The difference is at most $V(1/6 + \pi/32)|t|^3 e^{-Vt^2/\pi^2}$. Its integral divided by 2π, plus the Gaussian tail, is at most $(\pi^3/12 + \pi^4/64 + 1/\pi^2)/V < 6/V$. Integrating the product modulus gives $\sqrt{\pi}/(2\sqrt{2V}) < 1/\sqrt{V}$. For the last assertion the error is at most half the central Gaussian term, and $V \le m/4$ supplies the stated factor. □

A.3 Conditioning an independent product on a count

Lemma A.2. *Let (B_i, R_i) be independent pairs, with $B_i \sim \operatorname{Bern}(r_i)$, $r_i \in [a, 1-a]$, $\mathbb{E}R_i = 0$, $|R_i| \le d_0$. Allow additional independent centered Z_j with $|Z_j| \le d_0$. Put*

$$K = \sum_i B_i, \quad R = \sum_i R_i + \sum_j Z_j, \quad \mathbb{E}K = k \in \mathbb{Z}, \quad V = \operatorname{Var} K \ge 1000, \quad L = \operatorname{Var} R > 0,$$

and assume $\operatorname{Cov}(K, R) = 0$. Set $\beta = a(1-a)$. If g is continuous and integrable, with integrable Fourier transform supported on $[-T, T]$, and $d_0 T \le \beta/162$, then

$$\begin{aligned} &\sup_y |\mathbb{E}[g(R - y) \mid K = k] - \mathbb{E}g(\sqrt{L}Z - y)| \\ &\qquad \le \frac{\|g\|_1}{\sqrt{L}} \left\{ \frac{32}{\sqrt{V}} + \frac{8d_0}{\sqrt{L}} + 2T\sqrt{VL}e^{-\beta m/162} \right\}, \qquad Z \sim N(0,1). \end{aligned} \tag{A.5}$$

Proof. Write $\chi(s,t) = \mathbb{E}e^{is(K-k)+itR}$. For $|s| \le 1/4$, $|t| \le T$, each centered individual phase X is bounded by $1/2$. If $v = \mathbb{E}X^2$ and $b = \mathbb{E}|X|^3$, symmetrization gives $|\mathbb{E}e^{iX}| \le e^{-v/3}$, and $v^2 \le b/2$

gives $|\mathbb{E}e^{iX} - e^{-v/2}| < b/4$. The total phase variance is exactly $Vs^2 + Lt^2$, by the zero covariance. The third absolute moments sum to at most $4(V|s|^3 + d_0L|t|^3)$. Telescoping therefore gives

$$|\chi(s,t) - e^{-Vs^2/2-Lt^2/2}| \le 2(V|s|^3 + d_0L|t|^3)e^{-(Vs^2+Lt^2)/3}.$$

Its double integral is at most $56(VL)^{-1/2}(V^{-1/2}+d_0L^{-1/2})$; use $\int e^{-u^2/3}du = \sqrt{3\pi}$, $\int |u|^3e^{-u^2/3}du = 9$. For $1/4 \le |s| \le \pi$, a Bernoulli factor is at most $e^{-2\beta/81} \le 1 - \beta/81$ because $\sin(1/8) > 1/9$. The residual changes it by at most $Td_0 \le \beta/162$, giving $|\chi(s,t)| \le e^{-\beta m/162}$. Its double integral is at most $4\pi Te^{-\beta m/162}$. The omitted Gaussian count tail has integral at most $8\sqrt{2\pi}(V\sqrt{L})^{-1}e^{-V/32}$.

Fourier inversion in the count and residual uses $|\widehat{g}| \le \|g\|_1$. By Lemma A.1, $q = \mathbb{P}(K = k)$ differs from $q_0 = (2\pi V)^{-1/2}$ by at most $6/V$; $V \ge 1000 > 288\pi$ ensures $q \ge q_0/2$. Division by q bounds the three preceding errors by

$$\frac{8\|g\|_1}{\sqrt{L}}(V^{-1/2} + d_0L^{-1/2}), \quad 2\|g\|_1T\sqrt{V}e^{-\beta m/162}, \quad \frac{3\|g\|_1}{\sqrt{VL}}e^{-V/32}.$$

Replacing q_0/q by one in the Gaussian expectation costs at most $12\|g\|_1/\sqrt{VL}$. Enlarge $8+3+12$ to 32. The factors involving y have modulus one, proving uniformity. □

Lemma A.3 (An interval minorant)**.** *For an interval (a,b) put $w_0 = (a+b)/2$, $d = (b-a)/2$. Fix $L_0 > 0$, $C_0 \ge 0$, and choose*

$$A_g \ge C_0 + (|w_0| + d/2)/\sqrt{L_0}, \quad \omega = (2/d)e^{A_g^2/6}, \quad T = 4\omega.$$

Then

$$g(w) = \left(1 - \frac{(w-w_0)^2}{d^2}\right)\operatorname{sinc}\big(\omega(w-w_0)\big)^4$$

is at most $\mathbf{1}_{(a,b)}$, has integrable Fourier transform supported on $[-T,T]$, and satisfies

$$\|g\|_1 \le 4/\omega, \qquad \mathbb{E}g(\sqrt{L}Z - y) \ge \frac{\phi(A_g)}{4\omega\sqrt{L}} \quad (L \ge L_0,\ |y| \le C_0\sqrt{L}).$$

If the bracket in (A.5) is at most $\phi(A_g)/32$, the corresponding conditional interval probability is at least $\phi(A_g)/(8\omega\sqrt{L})$.

Proof. On $|w - w_0| \le 1/\omega$, the quadratic factor is at least $3/4$, sinc $\ge 5/6$, and hence $g \ge 1/4$. Its Gaussian density there is at least $\phi(A_g)/\sqrt{L}$. The negative part has integral at most $2/(d^3\omega^4)$, with density at most $\phi(0)/\sqrt{L}$. Since $\omega^3 = 8\phi(0)/(d^3\phi(A_g))$, subtraction leaves the asserted lower bound. The integrals of $\min(1, |\omega s|^{-4})$ and $s^2\min(1, |\omega s|^{-4})$ are $8/(3\omega)$ and $8/(3\omega^3)$; $\omega d \ge 2$ proves the L^1 bound. The transform of the fourth power of sinc is a compactly supported cubic spline. Multiplication by the quadratic differentiates it twice, leaving an integrable transform on the same support. The final assertion follows from Lemma A.2. □

A.4 Jitter on the actual count lattice

He and Cheng [5, Lemma A.4] prove the iid jitter estimate. We use the cancellation $\sum_i v_i a_i = 0$ to control unequal gaps. Use the normalization of Section 5, and let f_i include all original coordinate characteristic functions. Put

$$g(t) = \sum_i (1 - |f_i(t)|^2), \qquad g_0(t) = 2V(1 - \cos t).$$

For $\max|a_i|, \varepsilon_0 \le 1/4$,

$$\begin{aligned} |g - g_0| &\le \Lambda t^2, \\ |g' - g_0'| &\le \Lambda(2|t| + 2t^2), \\ |g'' - g_0''| &\le \Lambda(2 + 8|t| + 4t^2). \end{aligned} \tag{A.6}$$

To prove these bounds, couple each actual symmetrized difference with $D_0 = B_i - B_i'$, writing it as $D_0 + E$, where $E = a_i D_0 + U_i - U_i'$. For an extra coordinate use $D_0 = 0$. All interpolation points lie in $[-2, 2]$, and $\sum \mathbb{E}E^2 = 2\Lambda$. Taylor-expand $1 - \cos(td)$, $d\sin(td)$, and $d^2\cos(td)$ in d. The terms involving U vanish conditionally to first order, and the remaining first-order terms cancel by $\sum_i v_i a_i = 0$. The second derivatives give exactly (A.6).

If $T \ge 10$ and $\xi(T+1)^2 \le 1/56$, each interval $I_j = [2\pi j - 1/4, 2\pi j + 1/4]$ meeting $[-T, T]$ has a unique minimum t_j of g, with

$$|t_j - 2\pi j| \le 4\Lambda(T+1)^2/V, \qquad \left|\prod_i f_i(t)\right| \le e^{-V(t-t_j)^2/4} \quad (t \in I_j).$$

Indeed $g'' \ge 27V/16 \ge V$ there, and the endpoint derivatives have opposite signs: their reference magnitudes exceed $V/3$ while the errors are at most $V/14$. Outside these intervals $g \ge V/25$, giving product modulus at most $e^{-V/50}$. For $j \ne 0$, the raw jitter multiplier $\operatorname{sinc}(t/2)$ vanishes at $2\pi j$ and has Lipschitz constant at most $1/4$. Using $|t| \ge 5|j|$, integrate its linear bound against the displayed Gaussian and sum the harmonic series. The result over both signs is

$$\sum_{j \ne 0} \int_{I_j \cap [-T,T]} \frac{|\operatorname{sinc}(t/2) \prod_i f_i(t)|}{|t|}\, dt \le (1 + \log T) \left\{ \frac{2}{5V} + \frac{4\sqrt{\pi}}{5} \frac{\Lambda(T+1)^2}{V\sqrt{V}} \right\}. \tag{A.7}$$

For $j \ge 1$ the summand is at most $1/(5jV) + \sqrt{\pi}|t_j - 2\pi j|/(10j\sqrt{V})$.

We now verify (5.11). Standardize by B and put $d = T_X/B^2 \in [1/4, 2]$. The Lyapunov ratio is d/B, and the support bounds satisfy the hypotheses of (A.3). Raw frequencies $|u| \le 1/128$ lie in that estimate's range; raw jitter width one corresponds to $h = 1/d \le 4$. Its integral is at most $300/B^2$, and after (A.1) and multiplication by $\sqrt{m}$ it costs at most $400/\sqrt{m}$. Equation (A.7) costs at most $(1 + \log T)\{m^{-1/2} + \xi(T+1)^2\}$ in this normalization. Between raw frequencies $1/128$ and $1/4$, the product is bounded by $e^{-B^2u^2/3}$. Outside the resonance intervals it is bounded by $e^{-V/50}$. The Gaussian comparison tail beginning at $B/128$ is at most

$$(32854/B^2 + 1/192)e^{-B^2/32768}.$$

Their total contribution after smoothing and scaling is below $60000(1 + \log T)\sqrt{m}e^{-m/262144}$, using $V \ge 3m/16$. Finally the cutoff BT costs at most $24\sqrt{m}/(BT) < 56/T$; the standardized signed comparison has Lipschitz constant below one. These are exactly the terms of (5.10).

B Quantitative confinement of the original supports

This appendix proves Proposition 4.1. Its first two lemmas turn a small lattice-moment deficit and a small smoothing loss into explicit distances. The remaining argument applies them to the contact equations and records the constants needed for the theorem.

B.1 The deficit in Esseen's moment inequality

We use a variance-weighted decomposition to quantify the deficit; compare He and Cheng [5, Lemma A.9] for stability of a single lattice law. Put $c = c_{\mathrm{E}}$, $p = p_{\mathrm{E}}$, and $\tau = \tau_{\mathrm{E}}$ within this subsection. Let Y be centered, nondegenerate, and supported on a translate of $h\mathbb{Z}$. Its oriented deficit is

$$\delta_Y = \frac{c\mathbb{E}|Y|^3 - \mathbb{E}Y^3 - 3h\mathbb{E}Y^2}{h\mathbb{E}Y^2}.$$

Write P_- for the law of $-Y$ on $(0,\infty)$, P_+ for the law of Y there, and $m = \mathbb{E}Y_+ = \mathbb{E}Y_- > 0$. The centered two-point law $P_{u,v}$ at $-u, v$ has masses $v/(u+v), u/(u+v)$. Direct integration gives

$$\mathrm{Law}(Y) = \mathbb{P}(Y=0)\delta_0 + \int P_{u,v}\,\lambda(du,dv), \qquad \lambda(du,dv) = \frac{u+v}{m}P_-(du)P_+(dv).$$

Its variance-weighted mixing measure $\omega(du,dv) = uv\lambda(du,dv)/\mathbb{E}Y^2$ is a probability measure. Set $n = (u+v)/h \in \{1,2,\dots\}$ and $a = u/(u+v)$. The identity

$$c\{a^2 + (1-a)^2\} - (1-2a) = 3 + 2c(a-p)^2$$

gives the exact formula

$$\delta_Y = \int \{3(n-1) + 2cn(a-p)^2\}\,d\omega. \tag{B.1}$$

This proves the oriented inequality and, by reflection, Esseen's inequality (1.6), including its equality case.

If $\delta_Y < 3$, some component has $n = 1$. Its two points are the unique nearest lattice points straddling zero, say $-ah, (1-a)h$; in particular this translate does not contain zero. Its variance weight is at least $1 - \delta_Y/3$. Every other component has variance at least $h^2a(1-a)$, so this variance weight is at most its ordinary mixture mass. Thus, for a centered law Q,

$$\begin{aligned} \mathrm{Law}(Y) &= (1-r_Y)\,\mathrm{Law}(h(B_a - a)) + r_Y Q, \quad r_Y \le \delta_Y/3, \\ |a-p| &\le \sqrt{\frac{\delta_Y}{2c(1-\delta_Y/3)}}, \quad h^2a(1-a) \le \mathbb{E}Y^2 \le \frac{h^2a(1-a)}{1-\delta_Y/3}. \end{aligned} \tag{B.2}$$

If $|Y| \le M$, coupling the mixture and then the two Bernoulli laws bounds their Wasserstein distance by

$$W_1(\mathrm{Law}(Y), \mathrm{Law}(h(B_p - p))) \le \frac{2M}{3}\delta_Y + 2h\sqrt{\frac{\delta_Y}{2c(1-\delta_Y/3)}}.$$

Here W_1 is the infimum of $\mathbb{E}|Y-Z|$ over couplings.

For a collection of centered h-lattice variables, use one common orientation and set

$$V = \sum_i \mathbb{E}Y_i^2, \quad B = \sum_i \mathbb{E}|Y_i|^3, \quad K = \sum_i \mathbb{E}Y_i^3, \quad \Delta_L = (cB - K - 3hV)/(hV).$$

Degenerate coordinates have zero variance weight. Applying the preceding bound on $\delta_{Y_i} \le 1$, the crude bound $M + h$ on its complement, and Cauchy–Schwarz gives

$$\frac{1}{V}\sum_i \mathbb{E}Y_i^2 W_1(\mathrm{Law}(Y_i), \mathrm{Law}(h(B_p - p))) \le (5M/3 + h)\Delta_L + h\sqrt{3/c}\sqrt{\Delta_L}. \tag{B.3}$$

There are also useful moment-ratio estimates. Put

$$a_D = \sqrt{(1 + D/3)D/(2c)}, \quad b_D = \tau D/3 + D/c + (2/c)a_D, \quad k_D = D/(3c) + 2a_D.$$

If $\Delta_L \le D$, then

$$\left|\frac{B}{hV} - \tau\right| \le b_D, \qquad \left|\frac{K}{hV} - \frac{1}{c}\right| \le k_D, \qquad \frac{B}{hV} \ge \frac{1}{2}. \tag{B.4}$$

Indeed (B.1) gives $\mathbb{E}_\omega(n-1) \le D/3$, $\mathbb{E}_\omega[n(a-p)^2] \le D/(2c)$, and $|\mathbb{E}_\omega[n(a-p)]| \le a_D$. Expand $n\{a^2 + (1-a)^2\}$ and $n(1-2a)$ about p. The last inequality follows directly from $n \ge 1$. All these formulas allow arbitrary variance weights, so they also hold when each sum is multiplied by a common positive factor.

B.2 Rounding at an approximate period

For the selected array put $Y_i = X_i/\ell$. Then $|Y_i| \le 8$, $\mathbb{E}Y_i^2 \le 9$, and

$$\ell^2 \sum_i \mathbb{E}Y_i^2 = \ell^2 \sum_i \mathbb{E}|Y_i|^3 = 1.$$

At a nonzero frequency u, take the phase of $\mathbb{E}e^{iuY_i}$, using any phase if the expectation vanishes. Round Y_i to the nearest point of the corresponding lattice of span $h = 2\pi/|u|$ and then center the rounded variable, calling it Q_i. The inequality $|v| \le (\pi/2)|e^{iv} - 1|$ for $|v| \le \pi$ gives

$$e := \ell^2 \sum_i \mathbb{E}|Y_i - Q_i|^2 \le h^2 g(u)/8, \qquad |Q_i| \le 8 + h. \tag{B.5}$$

In detail the uncentered squared rounding error is at most $(h^2/16)\mathbb{E}|e^{iuY_i} - e^{i\theta_i}|^2 = (h^2/8)(1 - |\mathbb{E}e^{iuY_i}|)$, which is no larger than the expression using $1 - |\mathbb{E}e^{iuY_i}|^2$. Centering decreases that error. If $M = 8 + h$, Cauchy–Schwarz yields

$$\begin{aligned} &\ell^2 \sum_i |\mathbb{E}Q_i^2 - \mathbb{E}Y_i^2| \le 2\sqrt{e} + e, \\ \ell^2 \sum_i |\mathbb{E}|Q_i|^3 - \mathbb{E}|Y_i|^3|, \quad &\ell^2 \sum_i |\mathbb{E}Q_i^3 - \mathbb{E}Y_i^3| \le (3M/2)\sqrt{e}(2 + \sqrt{e}). \end{aligned} \tag{B.6}$$

For the cubic bounds use $||x|^3 - |y|^3|, |x^3 - y^3| \le (3M/2)|x - y|(|x| + |y|)$. These bounds depend on total variance, not on the number of coordinates.

B.3 Comparing two small smoothing losses

Lemma B.1. *Let F be the CDF of a finite positive measure and let $J = F * \mathrm{Unif}[-h/2, h/2]$, $h > 0$. Suppose $J(x)$ differs from an affine function of slope $q > 0$ by at most ε on $[-D_0 - 2h, 3h]$. Put $H(a) = J(a + h/2) - F(a)$. If $0 \le d \le D_0$ and $H(0), H(-d) \le s$, then, for $0 < r_0 \le h/4$ and $K_0 = \lceil D_0/h \rceil$,*

$$\operatorname{dist}(d, h\mathbb{Z}) \le 3r_0 + \frac{4s + (5 + 4K_0)\varepsilon + 4K_0 h\varepsilon/r_0}{q}. \tag{B.7}$$

Proof. For the measure ν of F, write

$$w_a(t) = (1 - (t-a)/h)\mathbf{1}_{(a,a+h)}(t), \qquad H(a) = \int w_a(t)\nu(dt) \ge 0.$$

For $0 < d < h$, define the nonnegative trapezoid

$$T_d(t) = \frac{1}{h}\int_{h/2-d}^{h/2-d/2} \mathbf{1}_{\{|x-t|<h/2\}}\, dx.$$

Its integral against ν equals $J(h/2-d/2)-J(h/2-d)$. Splitting at $-d,-d/2,0,h-d,h-d/2,h$ gives the pointwise bound

$$\min\{1,2(h-d)/d\}\,T_d(t)\le w_0(t)+w_{-d}(t).$$

Integrating and applying the affine approximation to the two J values gives

$$q\min(d/2,h-d)\le H(0)+H(-d)+2\varepsilon. \tag{B.8}$$

The pointwise bound also holds at the endpoints, with the open intervals in w_a, so atoms require no continuity assumption on F.

To transfer over several periods, let $\overline{H}_{r_0}(a)=r_0^{-1}\int_a^{a+r_0}H(t)dt$. Integrating the definition of the uniform convolution gives

$$\begin{aligned}\overline{H}_{r_0}(a+h)-\overline{H}_{r_0}(a)=r_0^{-1}\int_a^{a+r_0}[J(t+3h/2)-J(t+h/2)]dt\\ -(h/r_0)[J(a+r_0+h/2)-J(a+h/2)].\end{aligned}$$

The affine terms cancel, so its magnitude is at most $2\varepsilon(1+h/r_0)$. Moreover $H(a+t)\le H(a)+(t/h)\nu((a+t,a+t+h))$ and the mass is at most $2(qh+\varepsilon)$, by the affine bound on J. Hence $\overline{H}_{r_0}(a)\le H(a)+qr_0+\varepsilon r_0/h$. Write $d=kh+d_0$, $0\le d_0<h$. Transfer from $-d$ to $-d_0$ using the preceding difference bound. If d_0 is within $2r_0$ of an endpoint, the conclusion is immediate. Otherwise choose a point in the averaging interval where H is at most its average and apply (B.8), using $H(0)\le s$. The threshold displacement is at most r_0; the resulting bound is (B.7). All arguments of J stay in the stated window. □

B.4 The scale and the maximizing threshold

We prove Proposition 4.1. Retain the notation of Section 2 for the actual selected maximizer. For scaled coordinates write

$$Y_i=X_i/\ell,\quad v_i=\mathbb{E}Y_i^2,\quad M_i=\mathbb{E}[Y_i|Y_i|],\quad \theta=\ell^{-1}\sum_i\mathbb{E}X_i^3,\quad |\theta|\le 1.$$

Let ρ,r,T,ε,h be as in Proposition 3.2. We give the error quantities explicitly so their later specialization requires no unspecified constants:

$$\begin{aligned}u&=\rho+\ell/2, & D&=16u+256r,\\ H_h&=72r+5b_D, & Z^2&=10(u+18r),\\ S_0&=2\rho+18r+460\ell^2, & A_0&=2\rho+24\{4\ell^2+(Z+32\ell)^2/2+\ell/5\},\\ \gamma&=300\sqrt{A_0}+11S_0+131A_0, & W_0&=21D+3\sqrt{D}+42r,\\ E_P&=14000(u+18r+\eta)+1100Z^2+4000\ell+4000\ell^2+1100\eta,\\ E_{\mathrm{end}}&=E_P+6100\gamma+700\gamma^2+500H_h.\end{aligned} \tag{B.9}$$

Here a_D,b_D,k_D are those in (B.4). The proof below uses only

$$\begin{gathered}r\le 10^{-6},\quad D\le 10^{-3},\quad A_0\le 9/64,\quad \gamma\le 1/100,\\ E_P\le 10^{-4},\quad E_{\mathrm{end}}\le 1/2000,\quad W_0\le 1/4.\end{gathered} \tag{B.10}$$

These will follow from (4.1) at the end.

The case $h = 0$ would give $R \le \phi(0)/6 + \rho < C_{\mathrm{E}}$ by (3.3), so $h > 0$. Initially $h \le 128\pi < 404$ and $g(2\pi/h) \le r^2/4$. Equations (B.5)–(B.6) give $\sqrt{e} \le 101r$, $|Q_i| \le 412$. If $V_Q = \ell^2 \sum \mathbb{E}Q_i^2$ and $B_Q = \ell^2 \sum \mathbb{E}|Q_i|^3$, then $V_Q \ge 1 - 202r$ and $B_Q \le 1 + 125000r$. Since $B_Q/(hV_Q) \ge 1/2$, it follows that

$$h \le 2(1 + 125000r)/(1 - 202r) < 5/2.$$

Apply the same rounding bounds with this improved span. Writing $K_Q = \ell^2 \sum \mathbb{E}Q_i^3$, we obtain

$$\sqrt{e} \le r, \quad |V_Q - 1| \le 3r, \quad |B_Q - 1| \le 32r, \quad |K_Q - \theta| \le 32r. \tag{B.11}$$

The lattice inequality in either orientation implies

$$3h + |\theta| \le c + 256r. \tag{B.12}$$

The moment-error allowance is $[32(c + 1) + 9h]r < 256r$.

Unsmoothing at the actual threshold z, with shift $h\ell/2$, gives

$$R \le \phi(z)\{h/2 + \theta(1 - z^2)/6\} + u. \tag{B.13}$$

The Gaussian increment has error at most $\phi(0)h^2\ell^2/8$, and the skew correction adds at most $h\ell^2/18$; together they are below $\ell^2/2$. Since $\phi(z)$ and $|(1 - z^2)\phi(z)|$ are at most $\phi(0)$,

$$3h + |\theta| > c - 16u, \qquad R \le C_{\mathrm{E}} + u + 18r. \tag{B.14}$$

It follows that $h > 3/2$. Orient the rounded variables by $\operatorname{sign}\theta$, arbitrarily if $\theta = 0$. Equations (B.11)–(B.14) bound their aggregate lattice deficit by D, its denominator hV_Q being greater than one. The moment-ratio estimates yield

$$|h - h_{\mathrm{E}}| \le H_h, \qquad \bigl||\theta| - 1/\sqrt{10}\bigr| \le 34r + 3k_D + H_h/6. \tag{B.15}$$

For $D \le 10^{-3}$, $a_D \le \sqrt{D}/3$, $b_D \le D/2 + \sqrt{D}/9$, and $k_D \le D/18 + 2\sqrt{D}/3$. Thus $H_h < 1/40$ and the second error is below $1/10$. Since $31/16 < h_{\mathrm{E}} < 41/21$, this proves

$$19/10 < h < 2, \qquad |\theta| > 1/5. \tag{B.16}$$

If $\theta < 0$, the envelope in (B.13) has its maximum at zero because $h > |\theta|$; it would imply $|\theta| < 128r + 8u = D/2$, a contradiction. Hence $\theta > 0$. Finally $C_{\mathrm{E}} < (C_{\mathrm{E}} + 18r)e^{-z^2/2} + u$, so

$$|z| \le Z, \qquad 0 \le K - C_{\mathrm{E}} \le u + 18r + \eta. \tag{B.17}$$

For this last bound use $-\log(1 - x) \le 2x$ for $x \le 1/2$.

B.5 Deletion with the same period and the support spacing

Deleting coordinate i subtracts at most ℓ^2 from g. Thus its high-frequency product bound is at most $e^{1/2} < 2$ times the full-row bound. Keep the original h for every deletion. Write $\sigma_i^2 = 1 - s_i \ge 3/4$, $\kappa_- = \sum_{j \ne i} \mathbb{E}X_j^3$, let J_i be the CDF of that deletion plus the same jitter, and put

$$G_i^*(x) = \Phi(x/\sigma_i) + \frac{\kappa_-}{6\sigma_i^3}(1 - (x/\sigma_i)^2)\phi(x/\sigma_i).$$

The logarithmic remainder in Appendix A remains $(49/3)\ell^2 t^4$. The low-frequency error is bounded by $\ell^2 e^{-t^2/5}\{(49/3)t^4 + t^6/72\}$, whose positive integral is below $206\ell^2$. The jitter-factor error is

below ℓ^2. For $a = \delta/\ell$, the Gaussian comparison tail is at most $4/(3a^2) + \ell/(2a) < 5500\ell^2$; use $\int_a^\infty t^2 e^{-3t^2/8} dt \le 3/a$. Also G_i^* is Lipschitz with constant below $1/2$. The weaker signed smoothing bound therefore gives

$$\|J_i - G_i^*\|_\infty \le 6000\ell^2 + 24r\ell/\delta^2 + 2\log(T/\delta)e^{-\varepsilon^2/(2\ell^2)} + 20\ell/T \le 2\rho\ell. \tag{B.18}$$

The last inequality uses the four allocations in (3.2).

At any original support point $y = \ell t$, the contact equality and (2.6) give

$$|G_i(z-y) - \Phi((z-y)/\sigma_i) - R\ell| \le 450\ell^3. \tag{B.19}$$

For the constant, $|t| \le 8$, $v_i, |M_i| \le 9$, $\mathbb{E}|Y_i|^3 \le 27$, and $K \le 23/50$. The Gaussian remainder contributes less than $51\ell^3$, the quadratic K term less than $51\ell^3$, and the remaining cubic, moment, and linear terms less than $348\ell^3$. Equations (B.12), (B.18), and (B.19) then imply

$$0 \le [J_i(z - y + h\ell/2) - G_i(z-y)]/\ell \le S_0. \tag{B.20}$$

The scale corrections cost at most $10\ell^2$ after division by ℓ: use $\sigma_i^{-1} - 1 \le s_i$, $\sigma_i^{-3} - 1 \le 3s_i$, and $|\kappa_-/\ell| \le \theta + 27\ell^2$.

Fix a support point x. The function $\ell^{-1}J_i(z - x + \ell a)$ differs from an affine function of slope $\phi(0)$ by at most A_0 on $[-16 - 2h, 3h]$. Indeed $|a| \le 24$, $|z - x + \ell a| \le Z + 32\ell$, and

$$|(G_i^*)'(w) - \phi(0)| \le 4\ell^2 + w^2/2 + \ell/5.$$

Integrate and use (B.18). Apply Lemma B.1 to the scaled positive measure, using (B.20), $q = \phi(0)$, $K_0 \le 11$, and $r_0 = \sqrt{A_0} \le h/4$. Its bound is at most $3\sqrt{A_0} + (8/3)(4S_0 + 49A_0 + 110\sqrt{A_0}) < \gamma$. Thus for every pair of support points of each original coordinate,

$$\operatorname{dist}((y-x)/\ell, h\mathbb{Z}) \le \gamma. \tag{B.21}$$

B.6 The support polynomial and the two classes

Define

$$P_i(t) = t^3 - c|t|^3 + (3c/2)t^2 + 3(cM_i - v_i)t.$$

Every point of the full scaled support satisfies

$$P_i(t) \ge -E_P. \tag{B.22}$$

To verify this with constants, divide (2.8) by $\phi(0)\ell^3/6$ and use (2.7). Its Taylor remainder divided by ℓ^3 is below 224ℓ. Furthermore $|\phi''(z) + \phi(0)| \le (3/2)\phi(0)Z^2$, since $\|\phi''''\|_\infty \le 3\phi(0)$ and $\phi'''(0) = 0$. The respective error bounds are

$$13184(K - C_{\mathrm{E}}), \quad 1092Z^2, \quad 3584\ell, \quad 3500\ell^2, \quad 1008\eta.$$

For the first, use $|-|t|^3 + (3/2)t^2 + 3M_i t| \le 824$ and $6/\phi(0) < 16$. The last two come from $6R\ell s_i^2 + \eta b_i + 4\eta\ell s_i$. Equation (B.17) makes their sum at most E_P.

For a nondegenerate coordinate let $-a, b$ be its support extremes. Centering gives $v_i \le ab$ and $|M_i| \le v_i$. If $a, b \ge 1/100$, divide the endpoint inequalities by a, b and add. The linear terms cancel, giving

$$(c+1)a^2 + (c-1)b^2 \le (3c/2)(a+b) + E_P(a+b)/(ab).$$

The left side is at least $3(a+b)^2$, because $(c^2-1)/(2c) = 3$. Therefore $a+b \le c/2 + E_P/(3ab) \le 25/8 + 1/3 < 7/2$. If $a < 1/100$ and $a+b \ge 7/2$, then $b > 3.49$ and

$$P_i(b) \le b^2[-(c-1)b + 3c/2 + 3(c-1)a] < -1,$$

contradicting (B.22). If $b < 1/100$, use $P_i(-a) \le a^2[-(c+1)a + 3c/2 + 3(c+1)b] < -1$. Every support diameter is consequently below $7/2$.

Partition coordinates according as their scaled support diameter is less than $h/2$ or at least $h/2$. In the first class, (B.21) implies diameter at most γ, so centering gives $|Y_i| \le \gamma$. In the second class the diameter differs from h by at most γ: zero and two multiples are excluded by (B.16) and the diameter bound. Every support point lies within 2γ of an extreme. Split at the midpoint, use its upper-cluster indicator B_i, and take conditional means. This gives exactly

$$Y_i = h_i(B_i - p_i) + U_i, \quad \mathbb{E}(U_i \mid B_i) = 0, \quad |h_i - h| \le 5\gamma, \quad |U_i| \le 2\gamma. \tag{B.23}$$

Both cluster probabilities are positive, because both extremes are support points and the clusters are separated.

We next bound those probabilities away from zero and one. On $[-8, 8]$, $|P_i'| < 1800$. Moving an extreme to its cluster mean costs at most 3600γ. Conditional centering gives $|M_i - M_i^0| \le \mathbb{E}U_i^2 \le 4\gamma^2$ and $v_i - v_i^0 = \mathbb{E}U_i^2 \le 4\gamma^2$, where the superscript denotes the pure two-point law. The first bound uses the Lipschitz derivative of $x|x|$ and also holds if a cluster crosses zero. Their polynomial contribution is below $700\gamma^2$. At a pure-law endpoint the polynomial is $h_i^3 C_3(p_i) + h_i^2 C_2(p_i)$, with $|C_3| < 29$, $|C_2| < 75/8$. Its gap derivative for gaps at most $9/4$ is below 500. Thus moving to the ideal gap h_{E} leaves both endpoint values at least $-E_{\mathrm{end}}$.

Put $t_0 = 3\sqrt{10}/2$. At its upper and lower endpoints those ideal-gap values are

$$h_{\mathrm{E}}^3(1-p)^2 A(p), \qquad h_{\mathrm{E}}^3 p^2 B(p),$$

where

$$A(p) = t_0 - (c-1) + 4(c-1)p - 6cp^2,$$
$$B(p) = t_0 - 3(c-1) + (8c-4)p - 6cp^2.$$

For $p \le 1/4$, B is increasing and $B(p) \le (\sqrt{10} - 17)/8 < -1$. If $p \in [1/100, 1/4]$, the lower endpoint is below $-6/10000 < -1/2000$, since $h_{\mathrm{E}}^3 > 6$. For $p \le 1/100$, use $A(0) = -p_{\mathrm{E}} < -2/5$, $A' \le 21$, so $A(p) < -19/100$. For $p \ge 3/4$, A is decreasing and $A(p) \le (\sqrt{10} - 49)/8 < -5$, excluding $[3/4, 99/100]$. For $p \ge 99/100$, $B(1) < -2$ and $|B'| \le 46$ give $B(p) < -1$. In view of $E_{\mathrm{end}} \le 1/2000$, all these cases are impossible. Hence every two-cluster probability lies in $[1/4, 3/4]$.

B.7 The extra coordinates and the parameter check

Lattice stability and the rounding coupling give

$$\sum_i \ell^2 v_i W_1(\mathrm{Law}(Y_i), \mathrm{Law}(h(B_{p_{\mathrm{E}}} - p_{\mathrm{E}}))) \le W_0. \tag{B.24}$$

For the rounded variables (B.3) is at most $(1+3r)(20D + 2\sqrt{D}) \le 21D + 3\sqrt{D}$. The change of variance weights has total absolute mass at most $3r$, and the distances are at most 13, costing $39r$. The original variance-weighted rounding distance is at most $3r$, by $v_i \le 9$ and the squared error bound $e \le r^2$. This proves (B.24) with the $42r$ allowance in W_0.

The target law has absolute first moment $2hp_{\mathrm{E}}q_{\mathrm{E}} > 9/10$. On the small-diameter class $|Y_i| \le \gamma$. Its total original variance V_s is therefore at most $W_0/(9/10-\gamma) \le 2W_0$. Each remaining coordinate has scaled variance between $1/2$ and 2, by (B.16), (B.23), and $p_i \in [1/4, 3/4]$. Their number m satisfies

$$\frac{1-2W_0}{2\ell^2} \le m \le \frac{2}{\ell^2}, \qquad m \ge 1/(4\ell^2). \tag{B.25}$$

Calling the small scaled coordinates Z_j,

$$\frac{\sum_j \operatorname{Var} Z_j}{mh^2} \le \frac{4W_0}{h^2(1-2W_0)} < 3W_0.$$

The relative gap errors in (B.23) are below 3γ, the residuals satisfy $|U_i|/h < 2\gamma$, and $|Z_j|/h < \gamma$. Thus the conclusion follows once $3\gamma, 3W_0 \le \lambda$ and $\ell \le 1/(2\sqrt{N})$.

We finish by checking these conditions and (B.10). From (3.2), $\ell \le \rho/48000$ and $r = \rho/196608$. For $\rho \le 10^{-6}$,

$$D \le 17\rho, \quad Z^2 \le 20\rho, \quad H_h \le 43\rho + 3\sqrt{\rho}, \quad S_0 \le 3\rho, \quad A_0 \le 500\rho,$$
$$\gamma \le 7000\sqrt{\rho}, \quad W_0 \le 14\sqrt{\rho}, \quad E_P \le 51000\rho + 15100\eta.$$

For $\rho \le 10^{-12}$ these imply

$$E_{\mathrm{end}} \le 5 \cdot 10^7 \sqrt{\rho} + 2 \cdot 10^4 \eta.$$

Before enlargement its ρ and $\sqrt{\rho}$ coefficients are at most 34300072500 and 42701500, which proves this last bound. Substitute $\rho = 10^{-26}\lambda^2$, $\eta = 10^{-10}\lambda$:

$$E_{\mathrm{end}} \le 7 \cdot 10^{-6}\lambda < 1/2000, \quad \gamma \le 7 \cdot 10^{-10}\lambda, \quad W_0 \le 1.4 \cdot 10^{-12}\lambda.$$

All the stated conditions follow, including $E_P \le 10^{-4}$. The representations and the variance estimates concern the original coordinates throughout, proving Proposition 4.1.

C Conditional moments and the one-sided comparison

This appendix proves the estimates used in Section 5. All conditioning is on the actual Bernoulli count. Dependence between a label and its residual is preserved.

C.1 Exponential tilting and projection

Retain $p_i \in [1/4, 3/4]$, $v_i = p_i(1-p_i)$, $\sum_i v_i a_i = 0$, and $\Lambda_A = \sum_i v_i a_i^2$. Suppose $m \ge 10^{12}$ and $|k-\mu| \le 6\sqrt{m}$. Tilt by $e^{\theta K_0}$ so that the count mean is k, putting

$$r_i = \frac{p_i e^{\theta}}{1 - p_i + p_i e^{\theta}}, \quad w_i = r_i(1-r_i), \quad V_\theta = \sum_i w_i.$$

For $|\theta| \le \log 2$, $r_i \in [1/8, 7/8]$ and $w_i \ge 7/64 > 1/10$. The count mean changes by more than $6\sqrt{m}$ in both directions before that boundary, so the tilt exists and

$$|\theta| \le 60/\sqrt{m}, \quad |w_i/v_i - 1| \le 2|\theta|, \quad |r_i - p_i - \theta v_i| \le \theta^2 v_i. \tag{C.1}$$

The conditional law given $K_0 = k$ is unchanged. Define

$$c_\theta = \frac{\sum_i w_i a_i}{V_\theta}, \quad b_i = a_i - c_\theta, \quad L = \sum_i b_i(B_i - r_i), \quad Q = \sum_i b_i^2, \quad Q_\theta = \sum_i w_i b_i^2.$$

On the count event, $A = m_k^0 + L$, where

$$|m_k^0| = \left|\sum_i a_i(r_i - p_i)\right| \le 1800\sqrt{\Lambda_A}/\sqrt{m}, \quad |Q_\theta - \Lambda_A| \le 180\Lambda_A/\sqrt{m}, \quad Q \le 16\Lambda_A. \tag{C.2}$$

For the second bound expand $Q_\theta = \sum_i w_i a_i^2 - (\sum_i w_i a_i)^2/V_\theta$. The errors are at most $2|\theta|\Lambda_A$ and $8\theta^2\Lambda_A$, together at most $3|\theta|\Lambda_A$. Also $\Lambda_A/2 \le Q_\theta \le 3\Lambda_A/2$; using $w_i \ge 7/64$ proves the last bound. If $\Lambda_A = 0$, all the gap estimates are immediate, so suppose it is positive.

C.2 Fourier moments with their residual factors

For $|t| \le 1/4$, put $z_i(t) = r_i e^{it}/(1 - r_i + r_i e^{it})$. The first centered complex cumulant is $c_{i,1} = z_i - r_i$; the next three are

$$c_{i,2} = z_i(1 - z_i), \quad c_{i,3} = z_i(1 - z_i)(1 - 2z_i), \quad c_{i,4} = z_i(1 - z_i)(1 - 6z_i(1 - z_i)).$$

Both z_i and $1 - z_i$ have modulus at most two. Differentiation gives

$$|c_{i,1} - iw_i t| \le 10t^2, \quad |c_{i,2} - w_i| \le 20|t|, \quad |c_{i,3}| \le 20, \quad |c_{i,4}| \le 100.$$

For $M_j = \sum_i b_i^j c_{i,j}$, the exact cancellation $\sum_i w_i b_i = 0$ yields

$$\begin{gathered} |M_1| \le 10\sqrt{mQ}t^2, \quad |M_2 - Q_\theta| \le 20Q|t|, \quad |M_2| \le 6Q, \\ |M_3| \le 20Q^{3/2}, \qquad |M_4| \le 100Q^2. \end{gathered} \tag{C.3}$$

The tilted count transform F_θ satisfies $|F_\theta(t)| \le e^{-mt^2/50}$ on $[-\pi, \pi]$. Lemma A.1 gives $\mathbb{P}_\theta(K_0 = k) \ge 1/(3\sqrt{m})$. After division by this probability, upper bounds for the Gaussian integrals of powers $0, 1, 2, 4, 8$ are respectively

$$7, \quad 25m^{-1/2}, \quad 175m^{-1}, \quad 13125m^{-2}, \quad 3 \cdot 10^8 m^{-4}. \tag{C.4}$$

These follow by scaling $t\sqrt{m}$ in $(2\pi)^{-1}\int e^{-mt^2/50}|t|^j dt$. On $1/4 \le |t| \le \pi$, expand the moment before factoring the product. A term of degree $j \le 4$ removes at most four Bernoulli factors; the remaining product is at most $e^{-m/810}$. The conditional tail contribution is therefore at most $3m^{(j+1)/2}Q^{j/2}e^{-m/810}$. For $j = 1, 2, 4$ and $m \ge 10^8$, these are at most $\sqrt{Q}/\sqrt{m}$, $Q/\sqrt{m}$, and Q^2. For instance $e^{-m/810} \le 10^{18}/m^5$ reduces the verification to decreasing powers of m.

The first, second, and fourth moment numerators are exactly

$$F_\theta M_1, \quad F_\theta(M_1^2 + M_2), \quad F_\theta(M_1^4 + 6M_1^2M_2 + 3M_2^2 + 4M_1M_3 + M_4).$$

Equations (C.3)–(C.4), with the tails just bounded, imply

$$\begin{gathered} |\mathbb{E}(L \mid K_0 = k)| \le 2000\sqrt{Q}/\sqrt{m}, \\ |\operatorname{Var}(L \mid K_0 = k) - Q_\theta| \le 2000Q/\sqrt{m}, \qquad \mathbb{E}(L^4 \mid K_0 = k) \le 2000Q^2. \end{gathered} \tag{C.5}$$

Here is numerical detail for these enlargements. The mean's central coefficient is 1750. The second moment errors are at most $500Q/\sqrt{m} + 1312500Q/m + 2Q/\sqrt{m}$, and the squared mean costs at most $4 \cdot 10^6 Q/m$. The five fourth-moment central coefficients, in the displayed order, are at most

$$3 \cdot 10^{12}/m^2, \quad 47250000/m, \quad 756, \quad 140000/\sqrt{m}, \quad 700,$$

all multiplying Q^2. Their sum and the tail are below $2000Q^2$. Centering raises the fourth-moment bound to $20000Q^2$. Combining (C.2) and (C.5) proves (5.5)–(5.6).

For completeness, the relative label estimate (5.7) also follows numerically. Under the tilt, deleting a Bernoulli changes count variance by at most $1/4$ and displaces the new threshold from its mean by at most one. The Gaussian mass ratio has logarithm bounded by $1/V_\theta$. The $6/V$ errors and a lower bound by half the centered Gaussian mass then give a deleted-count ratio error at most

$$4/V_\theta + 28/(\phi(0)\sqrt{V_\theta}) \le 300/\sqrt{m}.$$

Each tilted label probability differs relatively from its original probability by at most $2|\theta| \le 120/\sqrt{m}$. Multiplication gives $500/\sqrt{m}$ for $m \ge 10^{12}$. This is a relative bound, so arbitrarily small within-cluster noise variances remain controlled after conditioning.

C.3 Proof of the one-sided loss

We prove (5.14), following the clipping method of He and Cheng [5, Lemma 3.2 and equation (83)]. First suppose V_0 is centered and $|V_0| \le a = 1/10$. For $u \ge 0$, $\mathbb{E}(V_0)_+ \le u + a\mathbb{P}(V_0 > u)$ and $f(u) \ge 3u/4$ give $\mathbb{P}(V_0 > u) + f(u) \ge 3\mathbb{E}|V_0|/8$. If $u = -r$, $0 < r \le a$, put $t = \mathbb{P}(V_0 > -r)$. Centering gives $0 \le at - r(1-t)$, hence $r \le 2at$. Also $\mathbb{E}|V_0| = 2\mathbb{E}(V_0)_+ \le 2at$, so $t + f(-r) \ge t - 5r/4 \ge 3t/4 \ge 3\mathbb{E}|V_0|/8$. For $a < r \le 13/20$, the probability is one and $1 + f(-r) \ge 3/16 \ge 3\mathbb{E}|V_0|/8$.

For general centered Y, clip at $1/20$, obtaining T, and put $b = \mathbb{E}T$, $V_0 = T - b$, $M_4 = \mathbb{E}Y^4$. Then

$$|b| \le 8000M_4, \quad \mathbb{P}(T \ne Y) \le 160000M_4, \quad \mathbb{E}|V_0| \ge \mathbb{E}|Y| - 16000M_4.$$

If $M_4 \le 1/160000$, then $|b| \le 1/20$, $|V_0| \le 1/10$, and $|u-b| \le 13/20$. Apply the bounded case at $u-b$, and compare $f(u-b)$ with $f(u)$. The total error coefficient is $(3/8)16000 + 160000 + (5/4)8000 = 176000$. This proves the first inequality in (5.14). Apply it to $-Y$ and $-f(-\cdot)$ for the second. All events in this proof are strict; atoms at the threshold are included with the conventions needed in (5.16).

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